# fancytests

Maths and physics tests in LaTeX

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**Sommario**

fancytests is a LaTeX package for writing school tests in mathematics and physics. It replaces the test-related part of fancylessons, the author's previous package, with a smaller and faster one. Every command takes named options (`key=value`); there are no more round, angle or bar delimiters.

It provides: a test header (full or compact) with the boxes required for students with DSA; exercises and questions with automatic score totals; multiple-choice questions with shuffled options and an exported answer key; reproducible randomisation of variants, numbers, questions and exercises; reduced versions of a test from the same source; teacher copies with solutions; standard styles for function graphs and physics diagrams.

An optional compatibility layer compiles the tests written with fancylessons without changes.

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# 1 Getting started

## 1.1 About this package

fancylessons was the author's earlier LaTeX package, used both for lesson notes (theorem and definition boxes, colours, page styles) and for tests. Over time it grew to more than 4000 lines and about 70 loaded packages, which made every compilation slow. fancytests keeps only what is needed for tests, redesigned with a uniform `key=value` syntax and new features (answer key, shuffled questions, reduced versions, teacher copies).

The references to fancylessons in this manual (speed comparisons, the `legacy` option, the table of old and new commands in section 12) are there mainly for the author, to compare the old and the new version and to reuse past tests. Other readers can safely ignore them.

fancytests and this manual were written by Matteo Bramardi together with Claude, an AI model by Anthropic, which wrote and tested most of the code under the author's direction.

## 1.2 Installation

The package is a single file, `fancytests.sty`. Put it in the same folder as the document and load it in the preamble.

On Overleaf, the simplest way to keep it up to date is to link it from GitHub: *Upload* → *From external URL*, with the address

```
https://raw.githubusercontent.com/Bradwave/fancytests/main/fancytests.sty
```

The *Refresh* button of the linked file downloads the latest version. Replace `main` with a release tag (e.g. `v1.0`) to freeze a project on that version, which is advisable for tests already given.

**Note.** On Overleaf, use TeX Live 2023 or later (Menu → Settings → TeX Live version). Older versions lack some LaTeX3 functions used by the package.

## 1.3 A minimal test

```
\documentclass[10pt]{article}
\usepackage{fancytests}
\usepackage[T1]{fontenc}
\usepackage{lmodern}
\renewcommand*\familydefault{\sfdefault}
\geometry{a4paper, top=15mm, bottom=15mm, left=17.5mm, right=17.5mm}
\linespread{1.25}

\testsetup{class=4M, subject=math, dsa=rid}

\begin{document}
\testheader[date=20/03/2026, title={Logaritmi}]

\exercise Risolvi le seguenti equazioni.
\begin{parts}[cols=2]
  \itemx[pts=1] $3^x = 9$
  \itemx[pts=1] $\log_2 x = 3$
\end{parts}

\exercise Rispondi.
\begin{parts}
  \itemx[pts=1] Quale funzione è sempre crescente?
    \begin{choices}
      \correct $e^x$
      \choice  $e^{-x}$
      \choice  $\ln(-x)$
      \choice[fixed] Nessuna delle precedenti
    \end{choices}
\end{parts}

\vfill
\finalscore
\end{document}
```

The package already loads babel (Italian), geometry, amsmath, amssymb, mathtools, tabularx, booktabs, enumitem, xcolor, bm, cancel, tikz, and, unless disabled, pgfplots and siunitx. Do not load them again with different options.

## 1.4 Package options

Options are given when loading the package: `\usepackage[seed=42, answers]{fancytests}`.

| Key | Default | Meaning |
|---|---|---|
| subject | math | Default subject: math or physics. Sets the colour (blue or green) and the word in the title. |
| plots | true | Load pgfplots. With false the ftplot environment is not available, but loading is faster. |
| units | true | Load siunitx (for \qty, \num, ...), set up with the decimal comma. |
| seed | 1 | Seed of the random generator (section 8). 0 = a different seed at every compilation. |
| answers | off | Teacher copy: correct options highlighted, solution boxes printed. |
| nofigures | off | Fast draft: every TikZ picture and graph is replaced by an empty frame. |
| cache | off | Store figures and recompile them only when their code changes (package memoize). Needs shell escape, which is enabled on Overleaf. |
| legacy | off | Load the old fancylessons commands (section 12). |

### 1.5 Compilation speed

On a typical test the package loads in about $0.8\,\mathrm{s}$, against $1.9\,\mathrm{s}$ for fancylessons. Most of the remaining time is spent drawing graphs. To speed things up:

- while writing, use nofigures: pictures become placeholders;
- use cache: each picture is drawn once and reused until its code changes. The first two compilations are slower, the following ones much faster;
- if a test has no graphs, use plots=false.

**Note.** With cache, a picture whose content depends on randomisation through a macro defined elsewhere (e.g. a shuffled graph stored in a macro) cannot be recognised as changed. After changing such pictures or the seed, use *Recompile from scratch* on Overleaf, which empties the cache.

## 2 Global settings: \testsetup

\testsetup{⟨*keys*⟩} sets options for everything that follows. In the preamble it sets the defaults of the document; the same keys can be given to \testheader for a single test.

| Key | Default | Meaning |
|---|---|---|
| subject | math | math / physics (also matematica, fisica); any other text is printed as given. |
| class | — | Class, e.g. 4M. |
| date | empty | Date; when empty, a blank __/__/____ is printed. |
| title | empty | Subtitle of the test (topic). |
| type | Verifica | Type of test: Verifica, Test, Interrogazione, ... |
| style | full | Header layout: full or compact. |
| titlebreak | auto | Compact header only: put the title on its own line when it does not fit next to the subject (auto), always (always) or never, letting it wrap (never). |
| dsa | none | DSA boxes: none, dsa, rid (section 3). |
| timing | true | Print start, end and hand-in time fields. |
| recovery | empty | Date of the original test, for a resit ("recupero del ..."). |
| gradebox | VOTO | Label of the grade box. |
| showversion | off | Print the copy number next to the class (#1, #2, ...). |
| reduced | off | Reduced version: questions marked noreduced are left out. |
| answers | off | Same as the package option, but can be switched on and off inside the document. |
| labels | box | Labels of multiple-choice options: box (□), letters (A, B, C, ...) or boxed (letters in small boxes). |
| mcq | none | Where multiple-choice questions are graded: none, band (answer table) or margin (section 7.4). |
| motivation | 0 | Space for the motivation after the options, in lines (0 = none). |
| motivation-style | blank | blank (empty space) or grid (light grey squares). |
| motivation-mark | below | Look of the motivation mark: line, below (line with a tiny M underneath) or label (tiny M before the line). |
| grid-frame | rules | Frame of the answer table: cells, rules or lines (section 7.4). |
| shuffle | true | Shuffle multiple-choice options. |
| shuffleparts | false | Shuffle the questions of every parts. |
| seed | 1 | Restart the random generator from this seed. |
| decimal | . | Decimal mark in the score boxes. |
| opacity | 12 | Intensity (%) of the coloured score boxes. |
| grade-min, grade-max | 2, 10 | Lowest and highest grade in the grade formula. |
| final | compact | Default style of \finalscore. |
| autoclose | true | Close automatically an exercise left open (section 4). |

The header keys also accept Italian names: materia, classe, data, titolo, tipo, orari, recupero, voto.

# 3 The header

\testheader[⟨*keys*⟩] starts a new test: it prints the header, resets exercise numbers and scores and increases the copy number. It accepts all the keys of \testsetup; the values given here hold for this test only.

## 3.1 Full header

**Example**

```
\testheader[class=4M, date=20/03/2026,
  title={Logaritmo, equazioni esponenziali e logaritmiche}]
```

Nome: . . . . . **Classe 4M**

Cognome: . . . . . 20/03/2026

Ora inizio: . . . . . → Ora fine: . . . . . Ora consegna: . . . . . .

**VOTO**

# Verifica di Matematica

Logaritmo, equazioni esponenziali e logaritmiche

## 3.2 Compact header

`style=compact` uses about half the height. The title and the class share the first row, names and times share the second, the DSA boxes and the resit date go on an optional third row. A thin rule closes the header. When the title does not fit on the first row it moves, as a whole, to a row of its own (key `titlebreak`).

**Example**

```
\testheader[style=compact, class=4M, subject=physics,
  title={Onde e acustica}, dsa=dsa, recovery=12/05/2026]
```

**Verifica di Fisica** Onde e acustica **Classe 4M** | __/__/_____

**VOTO**

Nome e cognome: . . . . . | Inizio: . . . . . → Fine: . . . . . | Consegna: . . . . . .

☐ Schemi e calcolatrice ☐ Riduzione es./tempo aggiuntivo ☐ Compensazione . . . . . recupero del 12/05/2026

## 3.3 DSA boxes, times, resits

- `dsa=dsa`: *Schemi e calcolatrice*, *Riduzione es./tempo aggiuntivo*, *Compensazione*;
- `dsa=rid`: *Uso della calcolatrice*, *Uso di schemi/mappe*, *Riduzione es./tempo aggiuntivo*;
- `dsa=none`: no boxes.

`timing=false` removes the time fields (e.g. for oral tests); `recovery=`⟨*date*⟩ prints "recupero del ⟨*date*⟩".

**Example**

```
\testheader[class=4G, type=Interrogazione, subject=physics,
  timing=false, date=02/02/2026]
```

Nome: . . . . . **Classe 4G**

Cognome: . . . . . 02/02/2026

**VOTO**

# Interrogazione di Fisica

# 4 Exercises

`\exercise[⟨keys⟩]` prints the heading "Esercizio $n$" followed by the text of the exercise.

| Key | Default | Meaning |
|---|---|---|
| `diff` | — | Difficulty from 1 to 3, shown as coloured squares. |
| `title` | — | A different heading, e.g. `title=Test intermedi`. `\exercise[Test intermedi]` also works. |
| `bonus` | off | Bonus exercise: heading "Esercizio Bonus*", points go to the bonus, not to the total. `\bonusexercise` is a shortcut. |
| `fixed`, `follow` | — | Used only when exercises are shuffled (section 8.4). |

## 4.1 Closing an exercise

`\exercisescore[⟨keys⟩]` closes the exercise and prints its score box on the right. Without options the score is the sum of the questions. Options: `pts` imposes the score (`\exercisescore[5]` is the same as `[pts=5]`), `label` changes the label (default E1, E2, ...), `short` makes the box narrower. For bonus exercises use `\bonusexercisescore`; `\bonusexercisenote` prints the explanatory note.

If an exercise is not closed, the next `\exercise` or `\finalscore` closes it (key `autoclose`). Close it explicitly before a `\newpage`, otherwise the score box ends up at the top of the next page.

**Example**

```
\exercise[diff=2] Un'onda si propaga con velocità $v = \qty{340}{m/s}$.
\begin{parts}
  \itemx[pts=1] Calcola la lunghezza d'onda per $f = \qty{170}{Hz}$.
  \itemx[pts=0.5] Come varia $\lambda$ se $f$ raddoppia?
\end{parts}
\exercisescore

\exercise Spiega la differenza tra scala lineare e logaritmica.
\exercisescore[2]

\bonusexercise Determina il tempo di dimezzamento dello Iodio-131.
\bonusexercisescore[0.5]
\bonusexercisenote
```

**Esercizio 1** Un'onda si propaga con velocità $v = 340\,\mathrm{m/s}$.

**a.** Calcola la lunghezza d'onda per $f = 170\,\mathrm{Hz}$. /1.0

**b.** Come varia $\lambda$ se $f$ raddoppia? /0.5

**E1:** /1.5

**Esercizio 2** Spiega la differenza tra scala lineare e logaritmica.

**E2:** /2.0

**Esercizio Bonus*** Determina il tempo di dimezzamento dello Iodio-131.

**EB:** /0.5

* L'esercizio bonus non è obbligatorio e non è necessario svolgerlo per raggiungere il voto massimo.

# 5 Questions: the `parts` environment

Questions are written inside `parts`, one `\itemx` each. Letters continue across several `parts` of the same exercise and restart at every `\exercise`. The score of each question appears at the right of its first line.

## 5.1 Options of `parts`

| Key | Default | Meaning |
|---|---|---|
| `stretch` | `1.5` | Spacing between questions, as a multiple of the line height (1 = no extra space). |
| `cols` | `1` | Number of columns; questions fill the rows from left to right. |
| `start` | — | Letter to start from (`start=3` starts at "c"). |
| `ragged` | off | Ragged-right text instead of justified. |
| `shuffle` | off | Shuffle the questions (section 8.4). |

## 5.2 Options of `\itemx`

| Key | Default | Meaning |
|---|---|---|
| `pts` | — | Points of the question; added to the exercise. |
| `diff` | — | Difficulty 1–3, squares at the left of the letter. |
| `mark` | off | Highlight the letter. |
| `upper` | off | Capital letter. |
| `noreduced` | off | Leave the question out of the reduced version. |
| `fixed`, `follow` | — | For shuffled questions. |

**Example**

```
\exercise Risolvi.
\begin{parts}[cols=2, stretch=1.3]
  \itemx[pts=1] $3^x - 2^x = 0$
  \itemx[pts=1] $\log_{10}(2x-1) + \log_{10}(x) = 1$
  \itemx[pts=1, diff=2] $e^{2x} - e^{x+1} = 0$
  \itemx[pts=1, mark] $5 \cdot 2^x - 12 = 2^x$
\end{parts}
```

**Esercizio 1** Risolvi.

**a.** $3^x - 2^x = 0$ /1.0

**b.** $\log_{10}(2x-1) + \log_{10}(x) = 1$ /1.0

**c.** $e^{2x} - e^{x+1} = 0$ /1.0

**d.** $5 \cdot 2^x - 12 = 2^x$ /1.0

## 5.3 Graphs and questions side by side

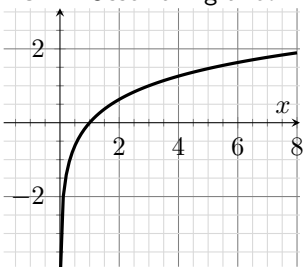
`parts` works inside a `minipage`; the scores stay at the right edge of the minipage.

**Example**

```
\exercise Osserva il grafico.

\begin{minipage}[c]{.36\linewidth}
  \centering
  \begin{ftplot}[xmin=-1.9, xmax=8.1, ymin=-3.9, ymax=3.1,
                 width=5.5cm, height=5cm]
    \fplot[domain=0.01:8]{ln(x)/ln(3)}
  \end{ftplot}
\end{minipage}\hfill
\begin{minipage}[c]{.6\linewidth}
  \begin{parts}[ragged]
    \itemx[pts=1] Quale funzione è rappresentata?
    \itemx[pts=1] Determina dominio e immagine.
  \end{parts}
\end{minipage}
```

**Esercizio 1** Osserva il grafico.




**a.** Quale funzione è rappresentata? /1.0

**b.** Determina dominio e immagine. /1.0

## 5.4 Reduced versions

Mark with `noreduced` the questions to leave out of the reduced (DSA) version, and compile with `\testsetup{reduced}`. The omitted questions disappear, the following letters move up and their points are not counted. Full and reduced versions come from the same file. Omitted questions do not appear in the answer key nor in the answer table; a `parts` whose questions are all omitted prints nothing.

**Example**

```
\testsetup{reduced}
\exercise Rispondi.
\begin{parts}
  \itemx[pts=1] Primo quesito.
  \itemx[pts=2, noreduced] Quesito omesso nella versione ridotta.
  \itemx[pts=1] Terzo quesito, stampato come b.
\end{parts}
\exercisescore
\testsetup{reduced=false}
```

**Esercizio 1** Rispondi.

**a.** Primo quesito. /1.0

**b.** Terzo quesito, stampato come b. /1.0

**E1:** /2.0

# 6 Scores and grade

## 6.1 Score commands

| Command | Meaning |
|---|---|
| `\score[⟨short⟩]{⟨n⟩}` | Add $n$ points to the current exercise and print the box. Inside `parts` use `pts` instead. |
| `\invscore{⟨n⟩}` | Add points without printing anything. |
| `\scorebox[⟨short⟩]{⟨n⟩}` | Print a box without adding points. |
| `\emptyscorebox` | An empty box. |
| `\exercisescore[⟨keys⟩]` | Close the exercise (section 4). |
| `\currenttotal` | Total of the closed exercises so far. |
| `\ResetAllScores` | Reset everything (done by `\testheader`). |

Scores are computed with exact decimal arithmetic and printed with up to two decimals: $1 \to 1.0$, $0.75 \to 0.75$.

## 6.2 Final score: `\finalscore`

`\finalscore[⟨keys⟩]` closes the last exercise, prints the summary and resets the scores, ready for the next copy.

| Key | Default | Meaning |
|---|---|---|
| `style` | `compact` | `table`: table E1, E2, ..., EB, Tot. and formula; `compact`: formula and total on one line; `formula`: formula only; `minimal`: total only; `none`. |
| `max` | sum | Maximum score shown, if different from the computed total. |
| `bonus` | `auto` | Bonus column in the table: `auto` (only if there are bonus points), `true`, `false`. |
| `label` | `Tot:` | Label of the total. |

The formula is voto $= \frac{\text{punti ottenuti}}{\text{punti massimi}} \cdot (\text{max} - \text{min}) + \text{min}$ with `grade-min` and `grade-max` of `\testsetup` (default $2$ and $10$).

**Example**

```
\exercise Primo. \begin{parts}\itemx[pts=2] a \itemx[pts=1.5] b\end{parts}
\exercise Secondo. \begin{parts}\itemx[pts=3] c\end{parts}
\bonusexercise Bonus. \bonusexercisescore[0.5]
\finalscore[style=table]
```

**Esercizio 1** Primo.

**a.** a /2.0

**b.** b /1.5

**E1:** /3.5

**Esercizio 2** Secondo.

**a.** c /3.0

**E2:** /3.0

**Esercizio Bonus *** Bonus.

**EB:** /0.5

| E1 | E2 | EB | Tot. |
|---|---|---|---|
| /3.5 | /3.0 | /0.5 | /6.5 |

$$\text{voto} = \frac{\text{punti ottenuti}}{\text{punti massimi}} \cdot 8 + 2 = \frac{}{6.5} \cdot 8 + 2 =$$

**Example**

```
\exercise Unico. \exercisescore[6]
\finalscore
```

**Esercizio 1** Unico.

**E1:** /6.0

$$\text{voto} = \frac{\text{punti ottenuti}}{\text{punti massimi}} \cdot 8 + 2 = \frac{}{6.0} \cdot 8 + 2 =$$

**Tot:** /6.0

# 7 Multiple-choice questions

## 7.1 The `choices` environment

Inside `choices`, `\correct` starts the correct option and `\choice` a wrong one. Options are shuffled; the number of columns is chosen automatically: all on one row if they fit, otherwise two columns, otherwise one. Long options wrap with a hanging indent.

**Example**

```
\exercise Rispondi.
\begin{parts}
  \itemx[pts=1] Quale funzione ha dominio $x \ge -3$?
    \begin{choices}
      \correct $\sqrt{x+3}$
      \choice  $\log(x+3)$
      \choice  $e^{x+3}$
      \choice  $\frac{1}{x+3}$
    \end{choices}
  \itemx[pts=1] Se $f(x)$ è una decrescita esponenziale, $\ln f(x)$ è
    \begin{choices}
      \correct una retta con pendenza negativa.
      \choice  una retta con pendenza positiva.
      \choice  una curva logaritmica decrescente.
      \choice[fixed] nessuna delle precedenti.
    \end{choices}
\end{parts}
```

**Esercizio 1** Rispondi.

**a.** Quale funzione ha dominio $x \geq -3$? /1.0

☐ $\frac{1}{x+3}$ ☐ $\sqrt{x+3}$ ☐ $e^{x+3}$ ☐ $\log(x+3)$

**b.** Se $f(x)$ è una decrescita esponenziale, $\ln f(x)$ è /1.0

☐ una retta con pendenza positiva. ☐ una retta con pendenza negativa.
☐ una curva logaritmica decrescente. ☐ nessuna delle precedenti.

## 7.2 Options

| Key | Default | Meaning |
|---|---|---|
| cols | auto | auto, or a fixed number of columns. |
| shuffle | true | Shuffle the options (global default in \testsetup). |
| labels | box | box, letters or boxed. |
| motivation | — | Motivation space for this question, in lines. |
| indent | 0pt | Left indent (\itmindent outside parts). |
| before | 2pt | Space above the options. |
| rowsep | 0pt | Extra space between rows of options. |
| after | .25em | Space below the options. |

Options of a single choice: \choice[fixed] keeps the option in its position (typically the last one, "Nessuna delle precedenti"); \correct[if={v=2}] makes the option correct only when the variant v chosen by \pick is the second (section 8).

Options can contain anything, including TikZ pictures.

## 7.3 Answer key and teacher copy

Every compilation writes the file ⟨*document name*⟩.key.csv, with one line per multiple-choice question:

```
versione;esercizio;quesito;risposta
1;2;b;C
1;2;c;A
2;2;b;B
```

`versione` is the copy number (one per `\testheader`), `esercizio` the exercise number (`B` for the bonus), `quesito` the letter of the question, `risposta` the letter of the correct option in the printed order. On Overleaf the file is in *Logs and output files*.

With the option `answers` (or `\testsetup{answers}`) the correct options are highlighted. Since the seed is the same, the teacher copy matches the student copies exactly.

**Example**

```
\testsetup{answers, labels=letters}
\exercise Rispondi.
\begin{parts}
  \itemx[pts=1] Capitale d'Italia?
    \begin{choices}
      \correct Roma \choice Milano \choice Torino \choice[fixed] Nessuna
    \end{choices}
\end{parts}
\begin{solution}
Roma è la capitale dal 1871.
\end{solution}
\testsetup{answers=false, labels=box}
```

**Esercizio 1** Rispondi.

**a.** Capitale d'Italia? /1.0

**A)** Roma **B)** Torino **C)** Milano **D)** Nessuna

**Soluzione**

Roma è la capitale dal 1871.

## 7.4 Answer table and motivations

When grading by hand, the answers, the students' motivations and the points should be in one place. A correct answer gets full points; the motivation is read only when the answer is wrong. Students mark each question on a short grey line ___ (the *motivation mark*) with a tick (✓) if they motivated the answer on the test, or with **F** if the motivation is on a separate sheet (*foglio a parte*). The library offers two layouts, chosen with `mcq`. The mark has three looks, chosen with `\testsetup{motivation-mark=...}`: `line` ___, `below` ___ (default) and `label` M ___.
M

**The answer table (`mcq=band`).** `\answergrid` prints a compact band of three rows: question labels, a cell for the answer letter with the motivation mark in its corner, and the score box, with the total at the end. Students tick the option next to the question and copy its letter into the table; you grade along one line and leave it only for wrong answers marked as motivated. With `mcq=band` the multiple-choice questions lose their score box next to the question: their points are in the table only (open questions keep theirs). The table lists every multiple-choice question of the copy, in printed order, so it follows shuffling and reduced versions.

**Example**

```
\testsetup{mcq=band}
\exercise Rispondi. Una risposta errata viene valutata solo se motivata.
\begin{parts}
  \itemx[pts=1] Quale funzione è sempre crescente?
    \begin{choices}\correct $e^x$ \choice $e^{-x}$ \choice $-\ln x$\end{choices}
  \itemx[pts=0.5] $\log_2 8$ vale
    \begin{choices}\correct 3 \choice 4 \choice 8 \choice 16\end{choices}
  \itemx[pts=2] Domanda aperta: dimostra che $\log_a b \cdot \log_b a = 1$.
\end{parts}
\answergrid
\testsetup{mcq=none}
```

**Esercizio 1** Rispondi. Una risposta errata viene valutata solo se motivata.

**a.** Quale funzione è sempre crescente?

☐ $e^{-x}$ ☐ $e^{x}$ ☐ $-\ln x$

**b.** $\log_2 8$ vale

☐ 8 ☐ 4 ☐ 16 ☐ 3

**c.** Domanda aperta: dimostra che $\log_a b \cdot \log_b a = 1$. /2.0

| | 1a | 1b | Tot |
|---|---|---|---|
| Risposta | M | M | |
| Punti | /1.0 | /0.5 | /1.5 |

M: ✓ motivata qui · F motivata su foglio a parte

The frame of the table is chosen with `\testsetup{grid-frame=...}` or `\answergrid[frame=...]`: `rules` (default, above) draws only the horizontal rules around the answers; `cells` frames each answer cell; `lines` draws no rules and gives each answer a writing line.

**Example**

```
\testsetup{mcq=band}
\exercise Rispondi.
\begin{parts}
  \itemx[pts=1] $\log_2 8$ vale
    \begin{choices}\correct 3 \choice 4 \choice 8 \choice 16\end{choices}
  \itemx[pts=1] $\log_3 1$ vale
    \begin{choices}\correct 0 \choice 1 \choice 3 \choice $-1$\end{choices}
\end{parts}
\answergrid[frame=cells, legend=false]
\answergrid[frame=lines, legend=false]
\testsetup{mcq=none}
```

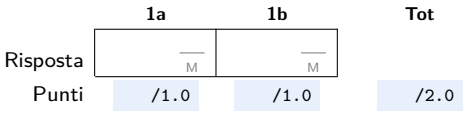
**Esercizio 1** Rispondi.

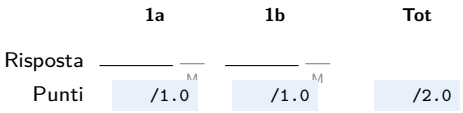
**a.** $\log_2 8$ vale

☐ 3 ☐ 8 ☐ 4 ☐ 16

**b.** $\log_3 1$ vale

☐ 3 ☐ 1 ☐ 0 ☐ $-1$




Options of `\answergrid`:

| Key | Default | Meaning |
|---|---|---|
| `style` | `band` | `band` (the table above) or `boxes` (one row per question: letter followed by A☐ B☐ . . . ). |
| `exercise` | `all` / current | Questions of one exercise only (default: whole copy for `band`, current exercise for `boxes`). |
| `cell` | `1.45cm` | Column width (`band`); about ten questions fit on one line, more continue on a second one. |
| `frame` | `grid-frame` | `cells`, `rules` or `lines` (`band`). |
| `legend` | `true` | Print the one-line legend of the motivation mark. |
| `cols` | 4 | Questions per row (`boxes`). |
| `title` | `Risposte:` | Title of the grid (`boxes`). |
| `before`, `after` | `.6em` | Space above and below. |

The questions are collected while the document is compiled and saved in the `.aux` file: if the table comes *before* the questions (e.g. on the first page), a second compilation is needed, which Overleaf runs by itself. The same holds for the first compilation with `mcq=band` or `mcq=margin`.

**The margin layout (`mcq=margin`).** No table: the motivation mark appears in the right margin, just left of the score box of each multiple-choice question, and the legend is printed once, after the first multiple-choice question of the copy.

**Example**

```
\testsetup{mcq=margin}
\exercise Rispondi.
\begin{parts}
  \itemx[pts=1] Quale funzione è sempre crescente?
    \begin{choices}\correct $e^x$ \choice $e^{-x}$ \choice $-\ln x$\end{choices}
\end{parts}
\testsetup{mcq=none}
```

**Esercizio 1** Rispondi.

**a.** Quale funzione è sempre crescente? /1.0 M

☐ $-\ln x$ ☐ $e^{-x}$ ☐ $e^{x}$

M: ✓ motivata qui · F motivata su foglio a parte

**Motivation space.** Off by default. `motivation=`⟨*lines*⟩ adds space after the options, as empty space or light grey squares (`motivation-style=grid`). It can be set for the whole test (`\testsetup`), for one exercise (`\exercise[motivation=2]`), for one question (`\itemx[motivation=3]`) or in `choices`.

**Example**

```
\testsetup{motivation-style=grid}
\exercise Rispondi motivando.
\begin{parts}
  \itemx[pts=1, motivation=2] $\log_2 8$ vale
    \begin{choices}\correct 3 \choice 4 \choice 8 \choice 16\end{choices}
\end{parts}
\testsetup{motivation-style=blank}
```

**Esercizio 1** Rispondi motivando.

**a.** $\log_2 8$ vale /1.0

☐ 16 ☐ 8 ☐ 3 ☐ 4

**Correction strips.** In the teacher copy (`answers`) the table shows the correct letters, and a last page collects one *correction strip* per copy: the correct letters with the same column widths as the student table, labelled with the copy number. Cut out the strip of a copy and lay it just under the student's answer row: each letter lines up with its answer.

## 7.5 Solutions

The `solution[`⟨*title*⟩`]` environment is printed only in the teacher copy (`answers`); otherwise it is ignored. The default title is *Soluzione*.

# 8 Randomisation

## 8.1 The seed

All random choices (shuffled options, questions and exercises, `\pick`, `\randint`, the old fancylessons commands) come from one generator. Its seed is fixed, so every compilation gives the same copies and the teacher copy and the answer key always match.

| Command | Meaning |
|---|---|
| `\usepackage [seed=42]{fancytests}` | Seed of the whole document (default 1). |
| `\testsetup{seed=42}` | Same; inside the document, it restarts the generator from that point. |
| `\setseed{42}` | Restart the generator from 42 here (e.g. before each class). |
| `seed=0` | A different seed at every compilation. |

**Note.** Do not use seed=0 for a real test: the answer key and the teacher copy would only match the student copies of the same compilation. To get new variants, change the number.

## 8.2 Variants and numbers

| Command | Meaning |
|---|---|
| `\pick[⟨name⟩]{a \or b \or c}` | One of the variants at random. With a name, the choice is remembered for `\correct[if={name=2}]`. |
| `\randint[⟨name⟩]{⟨min⟩}{⟨max⟩}` | A random integer between min and max, remembered with the name. |
| `\useval{⟨name⟩}` | Print again the number drawn with that name. |

**Example**

```
\exercise
Per quale funzione \pick[d]{$\Dom f = (1, +\infty)$ \or $\Dom f = (-1, +\infty)$}
si ha $\Ima f = \spR$?
\begin{choices}
  \correct[if={d=1}] $f(x) = -\ln(x - 1)$
  \correct[if={d=2}] $f(x) = -\ln(x + 1)$
  \choice $f(x) = -e^{x - 1}$
  \choice $f(x) = e^{x + 1}$
\end{choices}

Calcola $\randint[a]{2}{9} \cdot \randint[b]{2}{9}$.
Il risultato è $\useval{a} \cdot \useval{b} = \inteval{\useval{a}*\useval{b}}$.
```

**Esercizio 1** Per quale funzione $\operatorname{dom} f = (1, +\infty)$ si ha $\operatorname{im} f = \mathbb{R}$?

☐ $f(x) = -e^{x-1}$ ☐ $f(x) = -\ln(x-1)$ ☐ $f(x) = e^{x+1}$ ☐ $f(x) = -\ln(x+1)$

Calcola $4 \cdot 3$. Il risultato è $4 \cdot 3 = 12$.

## 8.3 Shuffling blocks

The `shuffle[⟨name⟩]` environment shuffles a list of blocks, which are then printed with `\shuffled{⟨name⟩}{⟨position⟩}`. Useful for graphs or pictures to be placed in a random order.

**Example**

```
\begin{shuffle}[g]
  \block{\tikz\draw[thick] (0,0) -- (1,1);}
  \block{\tikz\draw[thick] (0,1) -- (1,0);}
  \block{\tikz\draw[thick] (0,.5) -- (1,.5);}
\end{shuffle}
(a) \shuffled{g}{1} \qquad (b) \shuffled{g}{2} \qquad (c) \shuffled{g}{3}
```

(a)  (b)  (c) 

## 8.4 Shuffling questions and exercises

`parts[shuffle]` shuffles its questions (`\testsetup{shuffleparts}` does it for every `parts`). The environment `exercises` shuffles the exercises it contains. Letters and exercise numbers are assigned after shuffling; scores and the answer key follow.

- `fixed` keeps a question or exercise in its position (counted among the blocks: see below);
- `follow` ties it to the previous one, so they move together.

**Example**

```
\exercise Rispondi.
\begin{parts}[shuffle]
  \itemx[pts=1] Uno.
  \itemx[pts=1] Due.
  \itemx[pts=1] Tre: osserva il grafico.
  \itemx[pts=1, follow] Quattro: con riferimento al quesito precedente\dots
  \itemx[pts=1, fixed] Cinque (sempre nella quinta posizione tra i blocchi).
\end{parts}
```

**Esercizio 1** Rispondi.

**a.** Tre: osserva il grafico. /1.0

**b.** Quattro: con riferimento al quesito precedente... /1.0

**c.** Due. /1.0

**d.** Uno. /1.0

**e.** Cinque (sempre nella quinta posizione tra i blocchi). /1.0

```
\begin{exercises}
  \exercise ...            % each exercise runs until the next \exercise
  \exercise[fixed] ...     % stays in its place
  \exercise[follow] ...    % moves together with the previous one
\end{exercises}
```

Limitations:

- a question tied with `follow` forms a block with the previous one; `fixed` keeps the position among blocks, so letters after a block of two move down by one;
- shuffled questions must not refer to each other by letter;
- inside `exercises`, `\exercise` must not be inside a `minipage`; keep the bonus exercise outside the environment;
- `\verb` cannot be used inside `parts`.

## 8.5 Several copies

`\Repeat{`⟨*n*⟩`}{`⟨*code*⟩`}` repeats a test $n$ times. Each `\testheader` starts a new copy with new random choices; `showversion` prints the copy number, which is also the `versione` column of the answer key.

```
\newcommand{\verifica}{\input{docs/3-logaritmo/base}\newpage}
\Repeat{6}{\verifica}
```

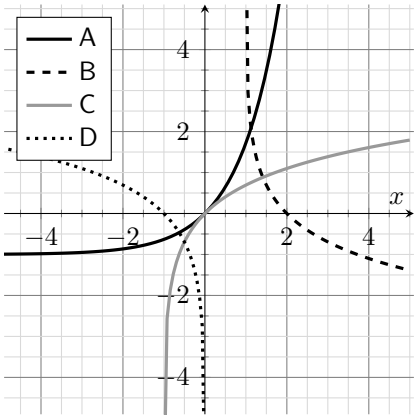
# 9 Graphs

The `ftplot[`⟨*axis options*⟩`]` environment draws a function graph with the standard style `ft`: centred axes, major and minor grid, thick black curves. Successive curves change style (solid, dashed, grey, dotted) so they can be told apart in black-and-white photocopies.

`\fplot[`⟨*options*⟩`]{`⟨*expression*⟩`}[`⟨*legend*⟩`]` adds a curve; the options are those of `\addplot` (`domain`, `samples`, `dashed`, ...). Any pgfplots option can be passed to `ftplot`.

**Example**

```
\begin{center}
\begin{ftplot}[xmin=-4.9, xmax=5.1, ymin=-4.9, ymax=5.1, width=7cm, height=7cm]
  \fplot[domain=-5:5]{e^x - 1}[A]
  \fplot[domain=1.001:5]{-ln(x - 1)}[B]
  \fplot[domain=-0.999:5]{ln(x + 1)}[C]
  \fplot[domain=-5:-0.001]{ln(-x)}[D]
\end{ftplot}
\end{center}
```




The style can also be used in an ordinary `axis`: `\begin{axis}[ft, xmode=log]`. The default number of samples is 80; increase it with `samples=150` for curves with sharp bends.

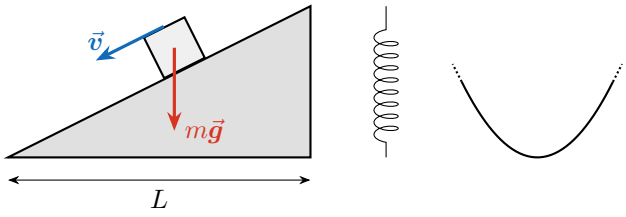
# 10 Physics diagrams

TikZ styles for physics diagrams:

| Key | Default | Meaning |
|---|---|---|
| `force` | — | Thick red arrow. |
| `velocity` | — | Thick blue arrow. |
| `vector` | — | Generic black arrow. |
| `axis arrow` | — | Thin arrow for axes. |
| `dimension` | — | Two-headed arrow for lengths. |
| `body` | — | Outlined body with light fill (for nodes). |
| `ground` | — | Grey fill for floors, slopes, walls. |
| `spring` | — | Spring (coil). |
| `continue`, `startcontinue`, `endcontinue` | `.75em` | Dotted continuation at both ends, at the start, at the end of a curve. |

**Example**

```
\begin{center}
\begin{tikzpicture}
  \fill[ground] (0,0) -- (4,0) -- (4,2) -- cycle;
  \draw[thick] (0,0) -- (4,0) -- (4,2) -- cycle;
  \node[body, minimum size=6mm, rotate=26.57] (b) at (2.2,1.45) {};
  \draw[force] (b.center) -- ++(0,-1.1) node[right] {$m\vb{g}$};
  \draw[velocity] (b.north) -- ++(-0.9,-0.45) node[above] {$\vb{v}$};
  \draw[dimension] (0,-.3) -- node[below] {$L$} (4,-.3);
  \draw[spring] (5,0) -- (5,2);
  \draw[thick, continue, domain=-1:1, samples=40] plot ({\x+7}, {\x*\x});
\end{tikzpicture}
\end{center}
```




# 11 Mathematics, units, text

| Command | Meaning |
|---|---|
| \spN, \spZ, \spQ, \spR[⟨*exp*⟩], \spC[⟨*exp*⟩] | $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{R}^2, \mathbb{C}$ |
| \Dom, \Ima, \Sign | $\operatorname{dom} f$, $\operatorname{im} f$, $\operatorname{sgn} x$ |
| \sen, \tg, \arcsen, \arctg | $\operatorname{sen} x$, $\operatorname{tg} x$, $\operatorname{arcsen} x$, $\operatorname{arctg} x$ |
| \abs{⟨*x*⟩}, \norm{⟨*x*⟩} | $\lvert x\rvert$, $\lVert x\rVert$ |
| \eqdef, \eqq | $\overset{\text{def}}{=}$, $\overset{?}{=}$ |
| \lesq, \gesq | $\leqslant$, $\geqslant$ |
| \vb{⟨*F*⟩}, \vect{⟨*AB*⟩} | $\vec{F}$, $\overrightarrow{AB}$ |
| \compacteq, \quitecompacteq, \notsocompacteq | Reduce the space around the next displayed equations (3, 6, 9 pt). |
| \unm{⟨*u*⟩}, \sunm[⟨*exp*⟩]{⟨*u*⟩}{⟨*v*⟩} | Light units: $3\,\mathrm{kg}$, $9.8\,\mathrm{m/s^2}$. For full units use siunitx: \qty{340}{m/s} gives $340\,\mathrm{m/s}$. |
| \tbf, \tit, \tbfit, \trm | Bold, italic, bold italic, roman. |
| \red, \bred, \blue, \bblue | red, **bold red**, blue, **bold blue**. |
| \hlc[⟨*colour*⟩]{⟨*text*⟩}, \highlight{⟨*text*⟩} | grey background , subject colour . |
| \dotline{⟨*length*⟩} | A dotted line to fill in: . . . . . |

Colours: MathsBlue, PhysicsGreen, SubjectColor (the colour of the current subject) and the colour-blind friendly palette FriendlyRed, FriendlyOrange, FriendlyYellow, FriendlyGreen, FriendlyForest, FriendlyBlue, FriendlyDenim, FriendlyMidnight.

# 12 Compatibility with fancylessons

With \usepackage[legacy]{fancytests} the tests written with fancylessons compile without changes. The old commands are read only when the option is given, so they do not slow down new tests. In legacy mode

exercises are not closed automatically, as in fancylessons.

| **fancylessons** | **fancytests** |
|---|---|
| `\mathtestheader[rec]{4M}{date}{title}(dsa)` | `\testheader[class=4M, date=..., title=..., dsa=dsa]` |
| `\physestestheader`, `\physinttestheader`, ... | `subject=physics`, `type=Interrogazione`, `timing=false` |
| `\begin {scoredlist}(1.5)<L> ... & \score {1}\\` | `\begin {parts}[ragged] ... \itemx[pts=1]` |
| `splitlist`, `threesplitlist` | `parts[cols=2]`, `[cols=3]` |
| `\randomfouroptions{a}{b}{c}{d}(.5)<L>\|0pt\|` | `\begin {choices} \correct a \choice b ...` |
| `\randomfouroptionsline`, `...list`, `...plusone...` | `choices[cols=4]`, `[cols=1]`, `\choice[fixed]` |
| `\randOp{a}{b}`, `\randOpThree`, `\randSign` | `\pick{a \or b}`, `\pick{+ \or -}` |
| `\randIntR{1}{9}`, `\randInt` | `\randint{1}{9}` |
| `\pgfmathdeclarerandomlist` + `\prunelist` | `shuffle` + `\shuffled` |
| `\minimalfinalscore` + `\finalscorenotable` | `\finalscore[style=compact]` |
| `\finalscorenotable` | `\finalscore[style=formula]` |
| `\overrideAllColors{`⟨*c*⟩`}` | `\colorlet{SubjectColor}{c}` |

Also available in legacy mode: `tflist`, `splittflist`, `tflistinv`, `\truefalse`, `\truefalseinv`, `\twooptions`, `\fouroptions`, `\fouroptionslist`, `\fiveoptionslist`, `\randomthreeoptionsline`, `\randomfiveoptions(line, list)`, `\randomfourplusoneoptionslist`, `\mathscore`, `\physscore`, `\intermediatetest`, `\lengthof`, `\nthof`, `notex*`, `\mathsol`, `\physsol` and the unit shortcuts `\unmeter`, `\unmpers`, ...

**Note.** The old option macros do not know which option is correct, so their questions do not appear in the answer key. To use the key, rewrite them with `choices` and `\correct`.

# 13 A complete test

```
\documentclass[10pt]{article}
\usepackage[seed=2026]{fancytests}
\usepackage[T1]{fontenc}
\usepackage{lmodern}
\renewcommand*\familydefault{\sfdefault}
\geometry{a4paper, top=15mm, bottom=15mm, left=17.5mm, right=17.5mm}
\linespread{1.25}

\testsetup{class=4M, dsa=none, style=compact, showversion}

\newcommand{\verifica}{%
  \testheader[date=20/03/2026, title={Logaritmi}]

  \exercise Risolvi le seguenti equazioni.
  \begin{parts}[cols=2, shuffle]
    \itemx[pts=1] $3^x - 2^x = 0$
    \itemx[pts=1] $\log_{10}(2x-1) + \log_{10}(x) = 1$
    \itemx[pts=1, noreduced] $2^x + 2 = 0$
    \itemx[pts=1, noreduced] $5 \cdot 2^x - 12 = 2^x$
  \end{parts}

  \exercise Rispondi.
  \begin{parts}
    \itemx[pts=1] Quale funzione è sempre crescente?
      \begin{choices}
        \correct $e^x$ \choice $e^{-x}$ \choice $-\ln x$
        \choice[fixed] Nessuna
      \end{choices}
  \end{parts}
  \begin{solution} $e^x$ è crescente perché la base è maggiore di 1. \end{solution}

  \vfill
  \finalscore
  \newpage}

\begin{document}
\Repeat{4}{\verifica}          % 4 different copies
% \testsetup{reduced}          % reduced (DSA) version
% \usepackage[answers]{...}    % teacher copy: same seed, same copies
\end{document}
```